

A Counterexample to the Conjectured Multiplicity at the Forced Points at Infinity

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agent_lane_08 (GPT5.6)

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Status: candidate_full_negative_solution_likely_valid.

The conjectured value is false. The sharp generic multiplicity is $\lceil (d \bmod n)/2 \rceil$, and this smaller value occurs inside the strictly hyperbolic locus for every residue covered by the conjecture.

Abstract

Simon and Vinzant conjecture that, when $r = d \bmod n \geq 3$, the two forced complex singularities $[0 : 1 : \pm i]$ of a C_n -invariant hyperbolic plane curve each have multiplicity $\binom{r}{2}$. We show that the invariant monomial support forces only the sharp lower bound $\lceil r/2 \rceil$ and that this is the generic exact multiplicity. For every $d = qn + r$ with $q \geq 1$ and $r \geq 3$, we construct normalized strictly hyperbolic real forms, invariant even under the associated dihedral symmetry, whose multiplicity at both points is exactly $\lceil r/2 \rceil$. Since $\lceil r/2 \rceil < \binom{r}{2}$, these are full counterexamples. In particular, for $(n, d) = (4, 7)$ the multiplicity is 2 rather than 3.

1 Source conjecture and evidence

The source is Faye Pasley Simon and Cynthia Vinzant, *Invariant hyperbolic curves: determinantal representations and applications to the numerical range*, arXiv:2102.01726 (2021) [1]. In Section 8.1, PDF/printed page 20, the authors write that for $d \bmod n \geq 3$ the invariant curves have complex singularities at $[t : x : y] = [0 : 1 : \pm i]$ and conjecture that each has multiplicity

$$\binom{d \bmod n}{2}.$$

They immediately recall the invariant monomial basis that we use below. The following is a direct crop rendered from that source page.

8. OPEN QUESTIONS AND FURTHER DIRECTIONS

8.1. Generalizing to Any Degree. One could hope to generalize Construction [4.9](#) for a hyperbolic plane curve of any degree. The main obstruction here is with assumption (A1), specifically the requirement that $\mathcal{V}_{\mathbb{C}}(f)$ is smooth. For curves with $d \bmod n \geq 3$, it seems there are always multiple singularities at infinity meaning most of these curves do not satisfy (A1). More specifically, there are complex singularities at the points $[t : x : y] = [0 : 1 : \pm i]$. We conjecture they each have multiplicity $\binom{d \bmod n}{2}$.

To see this recall that monomials in $t, x + iy$ and $x - iy$ form a basis for $\mathbb{C}[t, x, y]_d^{C_n}$, namely

$$(12) \quad \mathbb{C}[t, x, y]_d^{C_n} = \text{span}_{\mathbb{C}} \{ t^{d-j-k} (x + iy)^j (x - iy)^k : j \equiv k \pmod{n}, \text{ and } j + k \leq d \}.$$

The exponent vectors (j, k) are pictured in Figure [5](#).

Precise transcription. “For curves with $d \bmod n \geq 3$, it seems there are always multiple singularities at infinity [...]. More specifically, there are complex singularities at the points $[t : x : y] = [0 : 1 : \pm i]$. We conjecture they each have multiplicity $\binom{d \bmod n}{2}$.”

2 Definitions and local coordinates

Put

$$u = x + iy, \quad v = x - iy.$$

The degree- d invariant space has the basis recorded in the source:

$$\mathbb{C}[t, x, y]_d^{C_n} = \text{span}_{\mathbb{C}} \{t^{d-j-k} u^j v^k : j \equiv k \pmod{n}, j + k \leq d\}. \quad (1)$$

For a homogeneous plane curve $f = 0$, its multiplicity at a projective point is the least total degree of a nonzero term in a dehomogenized local Taylor expansion.

At $P_+ = [0 : 1 : i]$, work in the chart $x = 1$ and write $t = \tau$, $y = i + z$. Then

$$u = iz, \quad v = 2 - iz,$$

so v is a local unit. Consequently

$$\text{ord}_{P_+}(t^{d-j-k} u^j v^k) = (d - j - k) + j = d - k. \quad (2)$$

At $P_- = [0 : 1 : -i]$, the roles of u and v are reversed and the order is $d - j$.

3 Proof intuition

The source’s own exponent diagram contains the obstruction to the proposed number. At P_+ , maximizing the exponent k minimizes local order by (2). The congruence $j \equiv k \pmod{n}$ and the degree constraint allow k to reach $qn + \lfloor r/2 \rfloor$, not merely a value compatible with order $\binom{r}{2}$. The corresponding support point is unique, so its coefficient cannot cancel against another monomial. This gives the exact generic order $\lceil r/2 \rceil$.

To ensure this is not an artifact of leaving the hyperbolic locus, start with a radial form having d uniformly separated real roots in the t -direction and add the two conjugate extremal monomials. A compact sign argument keeps all d roots real and distinct for small perturbations, while the extremal term lowers the local multiplicity to $\lceil r/2 \rceil$ at both forced points.

4 The sharp support calculation

Write throughout

$$d = qn + r, \quad 0 \leq r < n, \quad s = \lfloor r/2 \rfloor, \quad \delta = r - 2s \in \{0, 1\}.$$

Lemma 1 (extremal invariant monomial). *Among the pairs (j, k) in (1),*

$$\max k = qn + s,$$

and equality holds only for $(j, k) = (s, qn + s)$. Symmetrically, $\max j = qn + s$, uniquely at $(j, k) = (qn + s, s)$.

Proof. Write $k = an + b$ with $0 \leq b < n$. Since $j \equiv k \pmod{n}$ and $j \geq 0$, we have $j \geq b$. Hence

$$d \geq j + k \geq an + 2b.$$

In particular $a \leq q$. If $a \leq q - 1$, then $k \leq (q - 1)n + (n - 1) = qn - 1 \leq qn + s$. If $a = q$, the displayed inequality gives $2b \leq r$, so $b \leq s$ and again $k \leq qn + s$.

The pair $(j, k) = (s, qn + s)$ is admissible because its exponent difference is qn and its sum is $qn + 2s \leq qn + r = d$. If $k = qn + s$, every nonnegative $j \equiv s \pmod{n}$ has the form $s + \ell n$; the degree constraint gives $\ell n \leq r - 2s = \delta < n$, hence $\ell = 0$. This proves uniqueness. The statement for j follows by symmetry. $\square$

Theorem 1 (sharp forced and generic multiplicity). *Every $f \in \mathbb{C}[t, x, y]_d^{C_n}$ vanishes at P_+ and P_- to order at least $\lceil r/2 \rceil$ whenever $r > 0$. This bound is sharp. More precisely, the coefficient of the unique extremal monomial*

$$t^\delta u^s v^{qn+s}$$

determines the lowest homogeneous part at P_+ , and the coefficient of its swapped monomial determines it at P_- . Thus a Zariski-open dense set of invariant forms has exact multiplicity $\lceil r/2 \rceil$ at both points.

Proof. By (2) and Lemma 1, every invariant basis monomial has order at least

$$d - (qn + s) = r - s = \lceil r/2 \rceil$$

at P_+ . Equality comes from the unique stated monomial. Therefore its nonzero coefficient gives exact order and cannot cancel. Swapping j, k proves the assertion at P_- . Nonvanishing of each of the two extremal coefficients is the complement of a pair of coefficient hyperplanes, hence is Zariski open and dense. $\square$

5 Strictly hyperbolic counterexamples

Recall that a real homogeneous degree- d form f is strictly hyperbolic with respect to $e = (1, 0, 0)$ if $f(e) > 0$ and, for every nonzero (x, y) , the polynomial $t \mapsto f(t, x, y)$ has d distinct real roots.

Theorem 2 (counterexamples for every conjectured residue). *Let $d = qn + r$ with $q \geq 1$ and $3 \leq r < n$. There exists $\varepsilon_0 > 0$ such that, for every real ε with $0 < |\varepsilon| < \varepsilon_0$, the form F_ε defined below is real, normalized by $F_\varepsilon(1, 0, 0) = 1$, strictly hyperbolic, and C_n -invariant (indeed invariant under swapping u, v), while*

$$\text{mult}_{P_+}(F_\varepsilon) = \text{mult}_{P_-}(F_\varepsilon) = \lceil r/2 \rceil < \binom{r}{2}.$$

Consequently the multiplicity conjecture in Section 8.1 of [1] is false.

Proof. Let $m = \lfloor d/2 \rfloor$ and define the radial form

$$H_d(t, x, y) = \begin{cases} \prod_{a=1}^m (t^2 - a^2 uv), & d = 2m, \\ t \prod_{a=1}^m (t^2 - a^2 uv), & d = 2m + 1. \end{cases}$$

Define the extremal real invariant form

$$G_{n,d} = t^\delta (u^s v^{qn+s} + u^{qn+s} v^s), \quad F_\varepsilon = H_d + \varepsilon G_{n,d}.$$

Both summands have degree d . Complex conjugation swaps u, v , so $G_{n,d}$ has real coefficients. Under the generator of C_n , u and v acquire reciprocal n th-root factors; the two exponent differences are $\pm qn$, so $G_{n,d}$ is invariant. It is also fixed by the reflection $u \leftrightarrow v$. The same properties are immediate for H_d , and $F_\varepsilon(1, 0, 0) = 1$.

At P_+ , the first monomial of $G_{n,d}$ has local order $\delta + s = \lceil r/2 \rceil$ and a nonzero initial term, while the second has larger order. At P_- their roles reverse. Each factor $t^2 - a^2 uv$ has local order 1 at either point, so H_d has order $\lceil d/2 \rceil$. Since $q \geq 1$ and $r < n$, we have $\lceil d/2 \rceil > \lceil r/2 \rceil$. Thus every $\varepsilon \neq 0$ gives exact multiplicity $\lceil r/2 \rceil$ at both points.

It remains to preserve strict hyperbolicity. On the unit circle $x^2 + y^2 = uv = 1$, $H_d(t, x, y)$ is the fixed monic polynomial

$$Q_d(t) = \begin{cases} \prod_{a=1}^m (t^2 - a^2), & d = 2m, \\ t \prod_{a=1}^m (t^2 - a^2), & d = 2m + 1, \end{cases}$$

whose d roots are distinct and real. Choose pairwise disjoint small closed intervals around those roots such that Q_d has opposite nonzero signs at the two endpoints of every interval. There are finitely many endpoints, and $G_{n,d}$ is continuous on their product with the unit circle. Hence, for all sufficiently small $|\varepsilon|$, the perturbation does not change any endpoint sign. The intermediate value theorem then supplies a real root of $F_\varepsilon(\cdot, x, y)$ in each of the d disjoint intervals. Because the polynomial is monic of degree d , these are all its roots and are distinct. Homogeneity reduces every nonzero (x, y) to the unit circle, proving strict hyperbolicity.

Finally, $\lceil r/2 \rceil < r(r-1)/2 = \binom{r}{2}$ for every $r \geq 3$, so the exact multiplicity contradicts the conjectured value. $\square$

The concrete $(n, d) = (4, 7)$ example

Here $q = 1$, $r = 3$, $s = 1$, and $\delta = 1$. The construction is

$$H = t(t^2 - uv)(t^2 - 4uv)(t^2 - 9uv), \quad G = t(uv^5 + u^5v), \quad F_\varepsilon = H + \varepsilon G.$$

At P_+ , using $u = iz$, $v = 2 - iz$, the initial part of G is $32i\tau z$, of total degree 2, whereas H begins in degree 4. At P_- the initial part is $-32i\tau z$. Thus both multiplicities are 2, not $\binom{3}{2} = 3$.

6 Verification, novelty, and limitations

Exact checks. The included script `code/verify_local_orders.py` expands the concrete example at both points, checks orders 4 and 2, verifies real coefficients and quarter-turn/reflection invariance, and brute-forces the support formula over a finite grid. These checks are not used as proof; the general argument is Lemma 1 and Theorems 1–2. Full commands and outputs are recorded in `verification.md`.

Bounded novelty search. On 27 August 2026, the run’s registry, solution, attempt, and proof-gap indexes were searched by arXiv id and core terms. Live web/arXiv searches used the exact conjecture sentence, title, authors, arXiv id, and combinations of “ $d \bmod n$ ”, “multiplicity”, “singularity”, “correction”, and “counterexample”. They found the source and mirrors, but no later paper or correction resolving this conjecture. This is a bounded search, not an exhaustive novelty guarantee.

Scope and interpretation. The argument uses ordinary plane-curve multiplicity, namely local Taylor order at the named projective points. It disproves both a universal reading and a generic reading: Theorem 1 gives the generic value, and Theorem 2 places it inside the strict hyperbolicity locus. No assertion is made here about the distinct determinantal-representation question that follows later in Section 8.1 of the source.

Human review recommendation. Check equation (2), the uniqueness clause in Lemma 1, and the compact endpoint-sign argument for strict hyperbolicity. No external result beyond the source's invariant basis is needed.

References

- [1] F. Pasley Simon and C. Vinzant, *Invariant hyperbolic curves: determinantal representations and applications to the numerical range*, arXiv:2102.01726, 2021.